

Amino Acids, Peptides, and Proteins

Biochemistry

Cline

September 25, 2025

Outline

3.1 Amino Acids

3.2 Peptides and Proteins

3.3 Working with Proteins

3.4 The Structure of Proteins: Primary Structure

Outline

3.1 Amino Acids

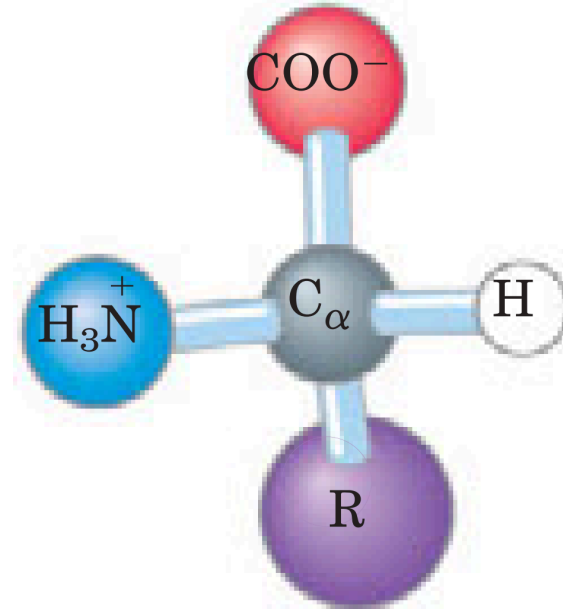
3.2 Peptides and Proteins

3.3 Working with Proteins

3.4 The Structure of Proteins: Primary Structure

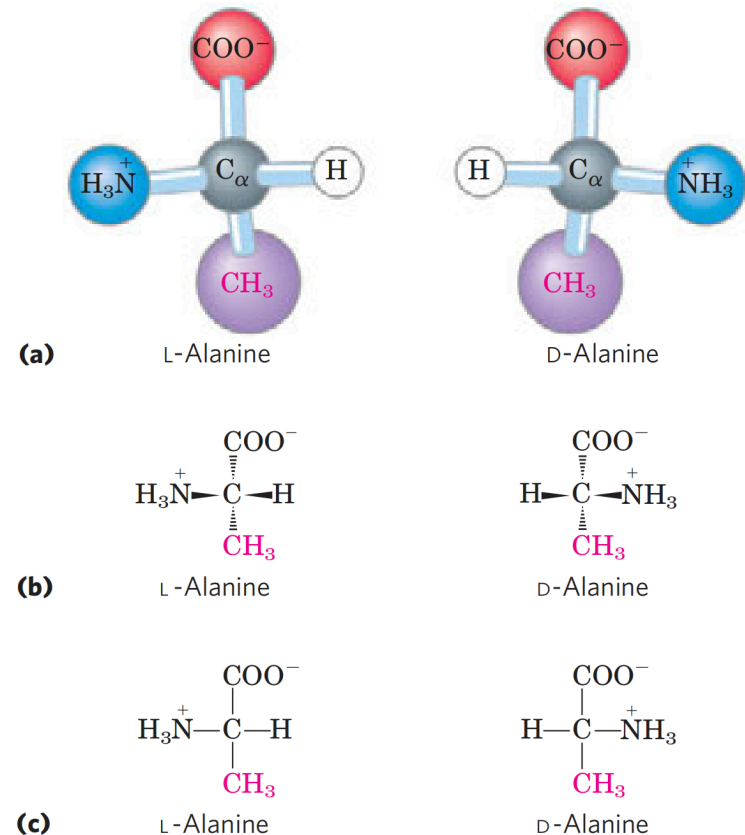
Common Structural Features

- Proteins are polymers of amino acids, joined by peptide bonds.
- All 20 common amino acids are α -amino acids.
- They have a carboxyl group and an amino group bonded to the α -carbon.
- They differ in their side chains (R groups).



Stereoisomerism

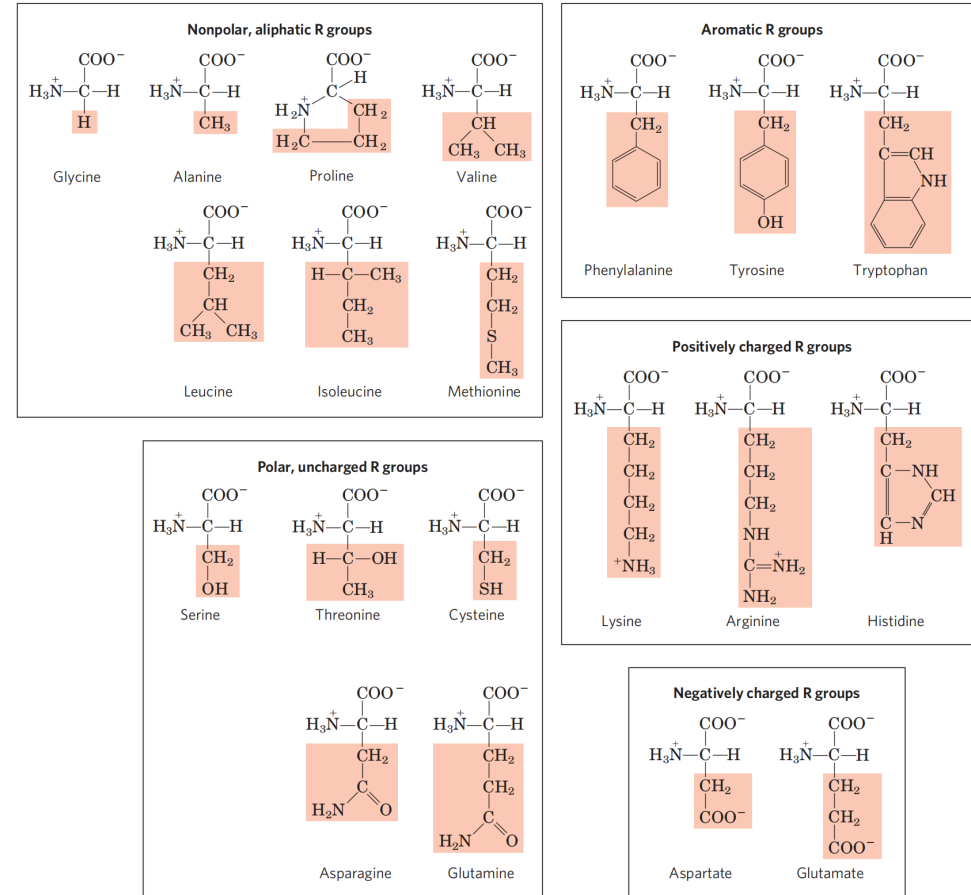
- For all amino acids except glycine, the α -carbon is a chiral center.
- This results in two possible stereoisomers: L and D forms.
- These forms are enantiomers (nonsuperposable mirror images).
- Nearly all amino acid residues in proteins are L stereoisomers.



Classification by R Group

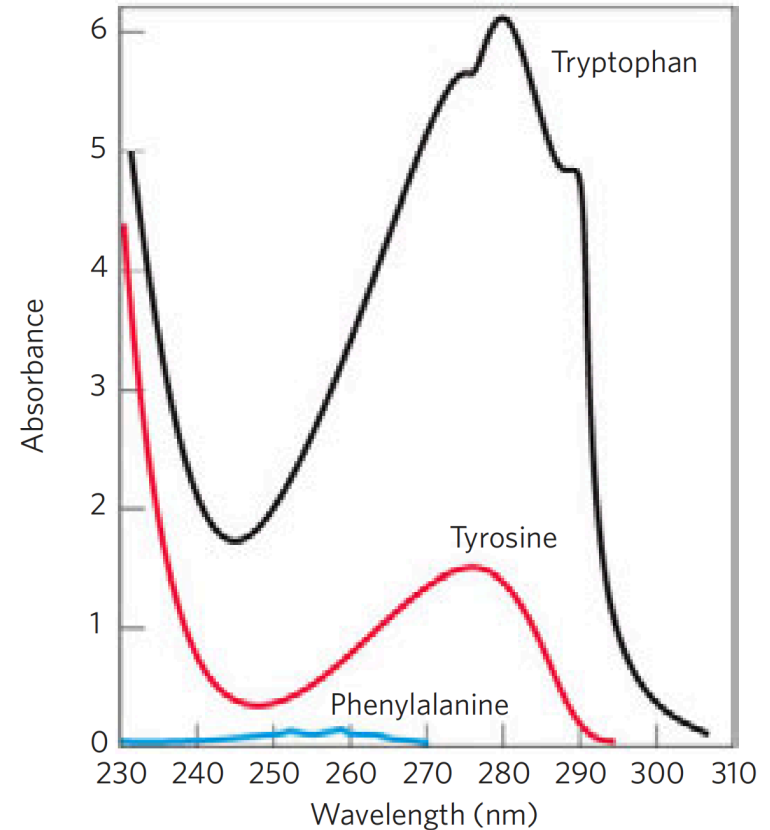
- Amino acids are classified into five groups based on the properties of their R groups:

- Nonpolar, aliphatic
- Aromatic
- Polar, uncharged
- Positively charged (basic)
- Negatively charged (acidic)



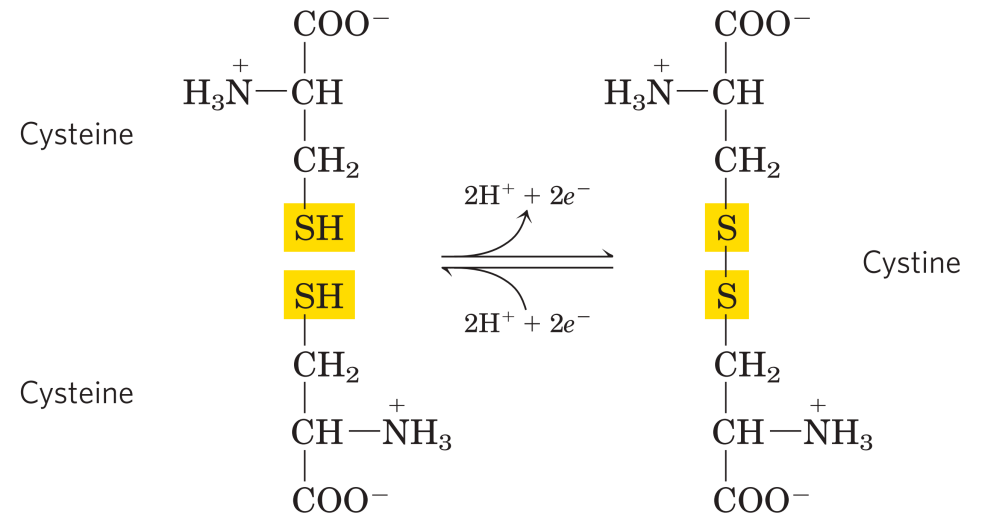
Aromatic R Groups & UV Absorption

- Phenylalanine, Tyrosine, and Tryptophan have aromatic side chains.
- They are relatively nonpolar (hydrophobic).
- Tyrosine and Tryptophan absorb UV light significantly at a wavelength of 280 nm, a property used to characterize proteins.



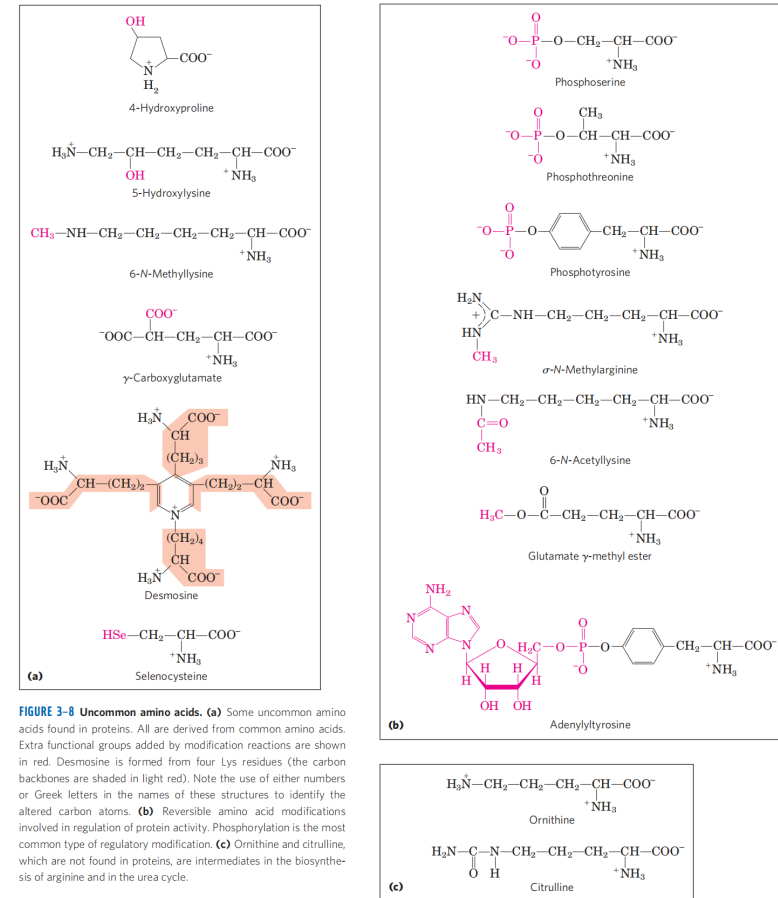
Polar, Uncharged R Groups & Disulfide Bonds

- Includes Serine, Threonine, Cysteine, Asparagine, and Glutamine.
- Their R groups are more hydrophilic and form hydrogen bonds.
- Cysteine can be oxidized to form a disulfide bond, creating a dimeric amino acid called cystine.
- Disulfide bonds stabilize protein structure.



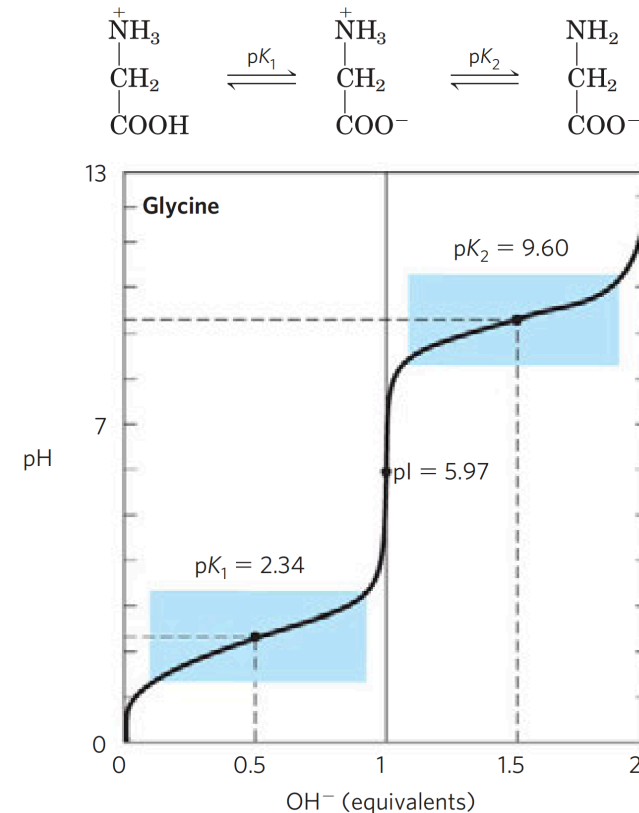
Uncommon Amino Acids

- Proteins can contain modified residues like 4-hydroxyproline and 5-hydroxylysine.
- Some residues are reversibly modified to regulate protein activity (e.g., phosphorylation).
- Other amino acids like ornithine and citrulline are key metabolic intermediates but not found in proteins.



Amino Acids as Acids and Bases

- At neutral pH, amino acids exist as zwitterions (dipolar ions).
- They have characteristic titration curves with specific pKa values for their carboxyl and amino groups.
- The isoelectric point (pI) is the pH at which the net charge is zero.



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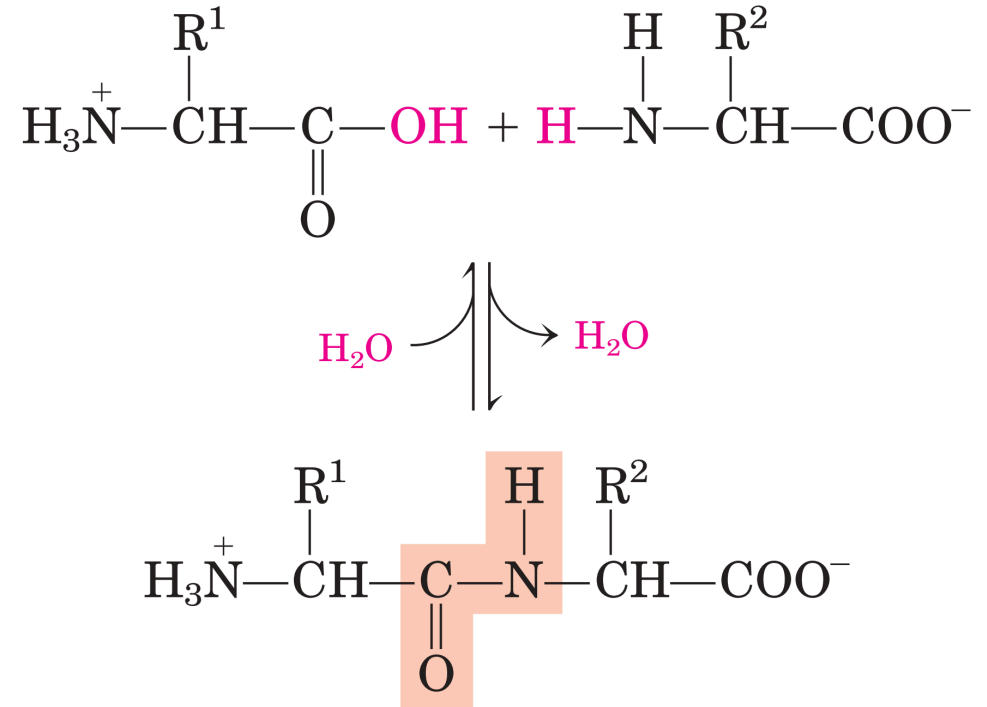
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Peptide Bond Formation

- Amino acids are joined by peptide bonds, a type of amide linkage.
- This is a condensation reaction where a water molecule is removed.
- A chain of amino acids is called a polypeptide.
- The ends are defined as the amino-terminus (N-terminus) and carboxyl-terminus (C-terminus).



Protein Composition

- Proteins can consist of a single polypeptide chain or multiple subunits (multisubunit proteins).
- **Simple proteins** contain only amino acids.
- **Conjugated proteins** contain a non-amino acid component called a prosthetic group (e.g., lipids, sugars, metals).

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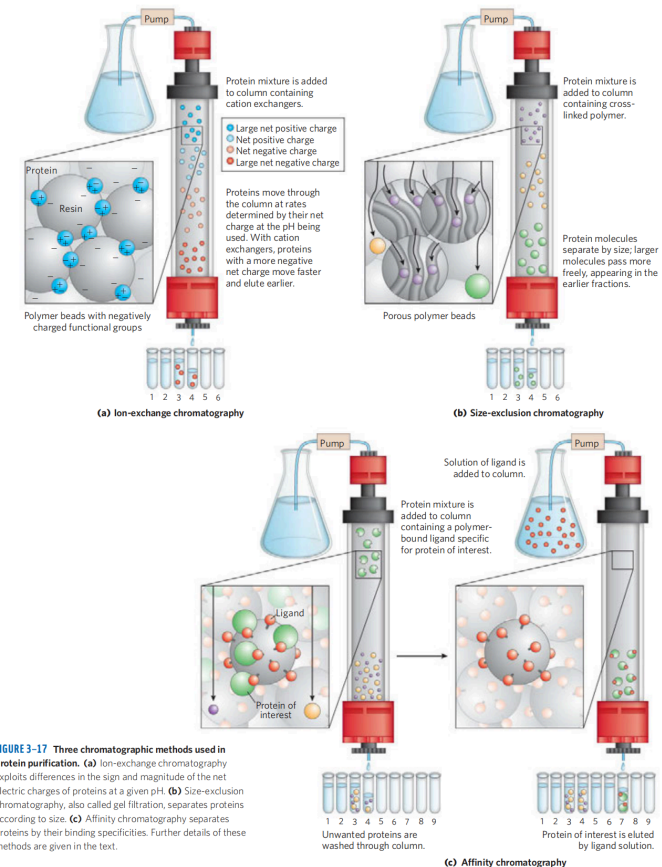
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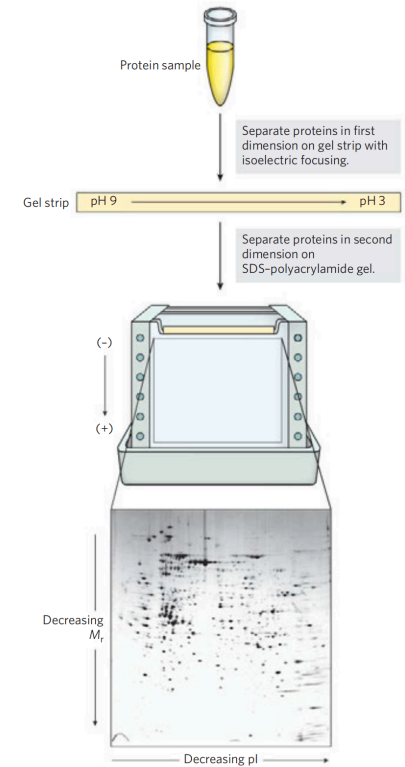
Protein Purification: Chromatography

- Proteins are purified from a crude extract using a series of fractionation steps.
- Column chromatography** is a key technique.
- Ion-exchange chromatography** separates based on net charge.
- Size-exclusion chromatography** separates based on size.
- Affinity chromatography** separates based on binding to a specific ligand.



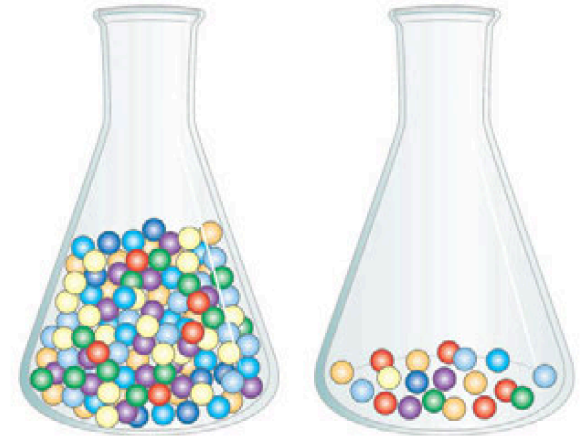
Protein Analysis: Electrophoresis

- **Electrophoresis** separates proteins based on the migration of charged proteins in an electric field.
- **SDS-PAGE** separates proteins by mass. SDS denatures proteins and provides a uniform negative charge.
- **Isoelectric focusing** separates proteins by their isoelectric point (pI).
- **Two-dimensional electrophoresis** combines both methods for high-resolution separation.



Quantifying Proteins

- For enzymes, the amount can be assayed by measuring catalytic activity.
- **Activity** refers to the total units of enzyme in a solution.
- **Specific activity** is the number of enzyme units per milligram of total protein.
- Specific activity is a measure of enzyme purity and increases during purification.



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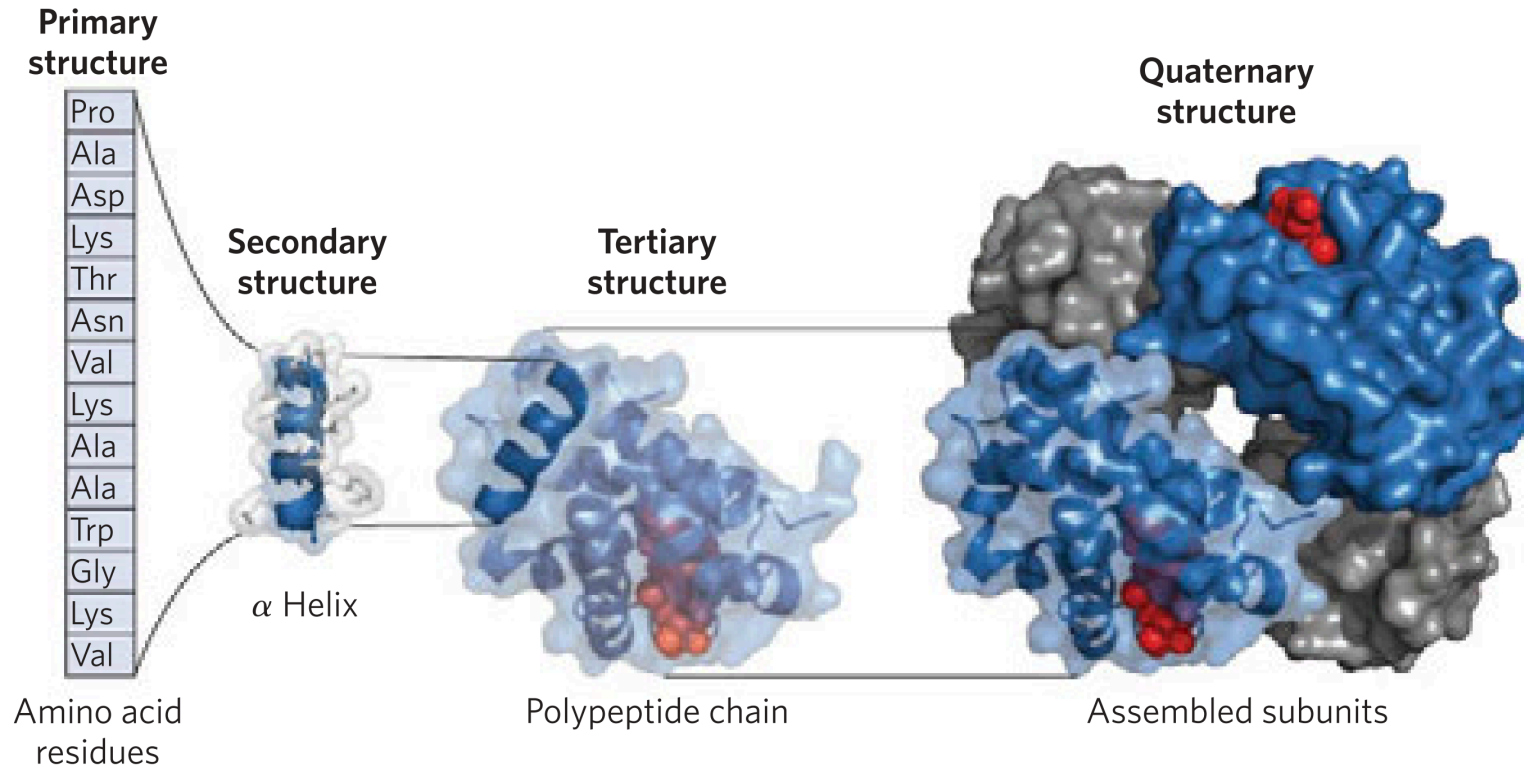
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Levels of Protein Structure

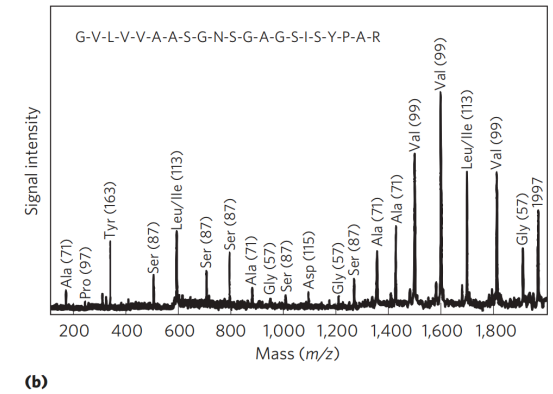
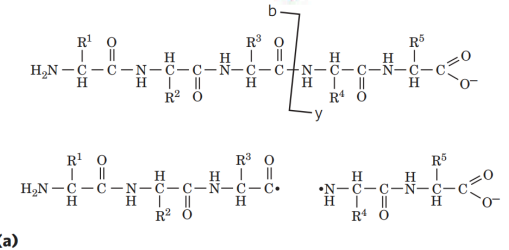
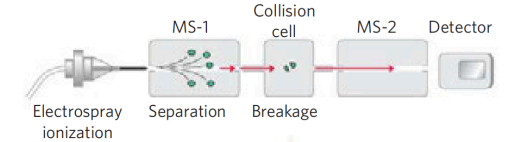


Levels of Protein Structure

- **Primary structure:** The sequence of amino acids.
- **Secondary structure:** Recurring structural patterns like the α -helix.
- **Tertiary structure:** The overall 3D folding of a polypeptide.
- **Quaternary structure:** The arrangement of multiple polypeptide subunits.

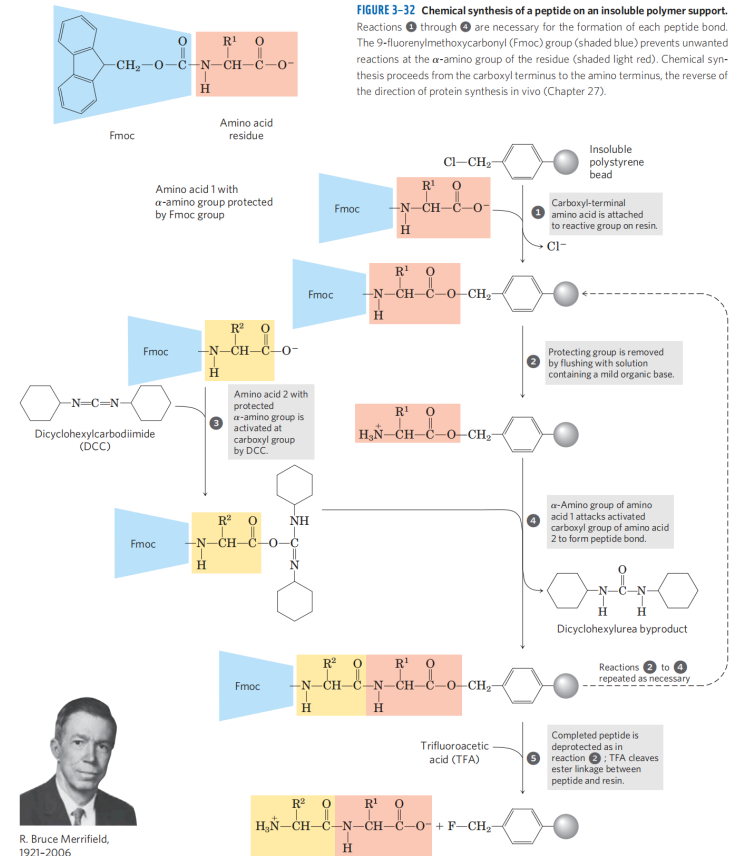
Determining Primary Structure

- The amino acid sequence determines the protein's 3D structure and function.
- Sequencing methods include:
 - **Edman Degradation:** Sequentially removes residues from the N-terminus.
 - **Mass Spectrometry:** Modern method (ESI-MS, MALDI-MS) that can sequence small amounts of protein quickly.



Chemical Peptide Synthesis

- Peptides can be chemically synthesized using solid-phase methods.
- The peptide is built one amino acid at a time while attached to an insoluble polymer support.
- This automated process is crucial for producing peptides for research and pharmacology.



Sequences and Evolution

- Comparing protein sequences helps understand evolutionary relationships.
- Homologous proteins** (homologs) share a common ancestor.
- Paralogs** are homologs in the same species.
- Orthologs** are homologs in different species.
- Evolutionary trees can be constructed based on sequence divergence.

